Synopsis

Heart Disease Prediction using Machine Learning Techniques

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- **<u>Problem Description:</u>**

Heart disease is a major global health concern and early prediction plays a critical role in reducing mortality rates. This project aims to develop an end-to-end machine learning classification system that predicts the presence of heart disease based on patient clinical data.

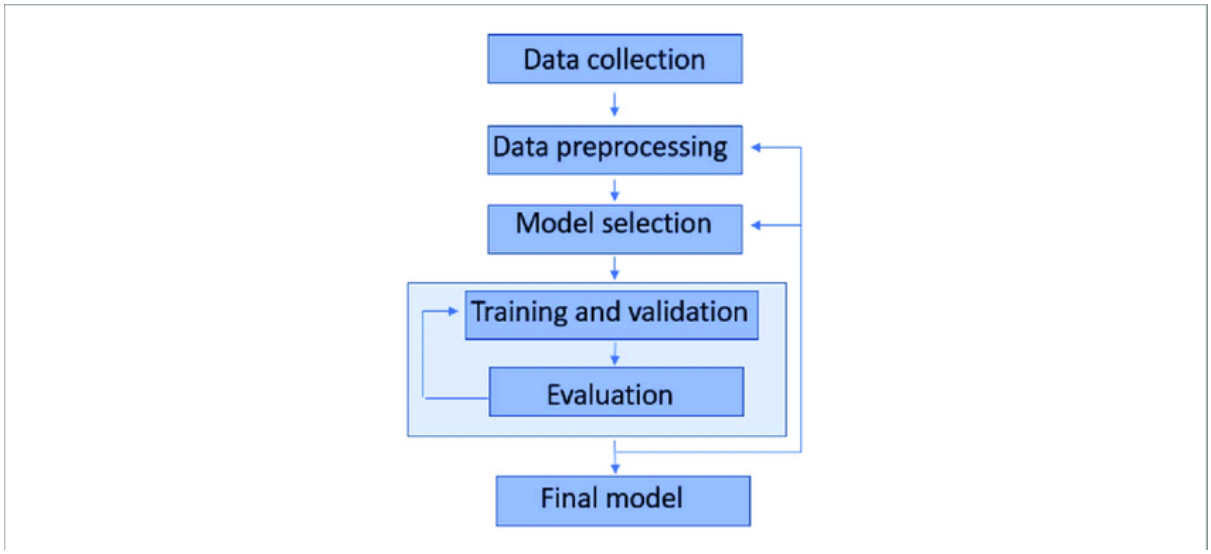
The dataset consists of features such as blood sugar level, depression, age, sex, blood pressure, and cholesterol levels. These attributes are essential indicators used by medical professionals to assess heart health.

The system will preprocess the dataset by handling missing values, normalizing and standardizing features, and transforming data into a suitable format. Machine learning models such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) will be applied to classify patients into two categories: presence or absence of heart disease.

The final objective is to build a reliable predictive system that can assist in early diagnosis and support healthcare decision-making.

- **<u>Steps of Implementation:</u>**

- **Dataset Design / Collection**
    - Dataset includes attributes:
      Patient ID, Patient Name, Age, Sex, Blood Pressure (BP), Cholesterol, Blood Sugar Level, Depression Assessment
- **Data Preprocessing**
    - Handling missing (NaN) values using Mean and Median
    - Data Cleaning
    - Normalization and Standardization of features
- **Exploratory Data Analysis (EDA)**
    - Understanding feature importance
    - Checking correlations between variables
- **Feature Engineering / Selection**
    - Selecting relevant clinical features for better accuracy
- **Model Training**
    - Logistic Regression
    - K-Nearest Neighbors (KNN)
    - Support Vector Machine (SVM)
- **Prediction System**
    - Classifies patient data into heart disease categories



- **<u>Expected Modules/ Libraries to be used:</u>**

  NumPy – Numerical computations

  Pandas – Dataset handling and manipulation

  Matplotlib / Seaborn – Visualization of data

  Scikit-learn – Machine learning models and preprocessing tools

- **<u>Expected Evaluation metrics and Plots to be used:</u>**
  - **Accuracy** – Overall correctness of prediction
  - **Precision** – Correctness of positive predictions
  - **Recall** – Ability to identify actual positives
  - **F1-Score** – Balance between precision and recall
  - **Confusion Matrix** – Visualization of prediction results
  - **ROC Curve** – Performance comparison across models

This helps evaluate model reliability, especially in medical classification tasks where accuracy alone is not sufficient.

- **<u>Required topics from the subject:</u>**
  Data Handling (Arrays & Strings)
  Used for storing and processing patient data
  File Handling
  Used for reading datasets and saving outputs
  Functions & Modules
  Helps in organizing code into reusable components

- **<u>Platform Required:</u>**

Visual studio Code

Jupyter Notebook

Python Idle.

- **<u>Books and Link Sources:</u>**
- Scikit-learn Documentation
- UCI Machine Learning Repository
- Kaggle Datasets
- Google Collab
- Youtube

- **<u>Expected Output:</u>**

  0 → No Heart Disease

  1 → Heart Disease Present